

Affine Invariance (1 + Bonus 0.5 pt)

(a) Show that Newton's method is invariant under affine transformations. Suppose $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a twice differentiable function. Consider the function $g(y) = f(Ay + b)$, where A is a non-singular constant matrix and $A \in \mathbb{R}^{n \times n}, b \in \mathbb{R}$. Prove that the sequence of points $\{x^k\}$ of f and the sequence of points $\{y^k\}$ of g generated by Newton's Method, starting from x^0 and y^0 respectively with $x^0 = Ay^0 + b$, have one-to-one correspondence under the transformation, i.e. $f(x^k) = g(y^k)$. Here we assume that the step lengths of the search directions of both functions are equal.

Let's first try to relate the gradient and Hessian of g in terms of f .

We know that $g(y) = f(x)$ with $x = Ay + b$

$$\text{Gradient: } \nabla g(y) = \begin{bmatrix} \partial g / \partial y_1 \\ \vdots \\ \partial g / \partial y_n \end{bmatrix} \quad \frac{\partial g}{\partial y_j} \underset{\substack{\uparrow \\ \text{Chain} \\ \text{rule}}}{=} \sum_{i=1}^n \frac{\partial f}{\partial x_i} \cdot \frac{\partial x_i}{\partial y_j} = \sum_{i=1}^n \frac{\partial f}{\partial x_i} \cdot A_{ij}$$

$$\text{Therefore } \nabla g(y) = A^T \nabla f(x)$$

$$\text{Hessian: } \nabla^2 g(y) = \begin{bmatrix} \frac{\partial^2 g}{\partial y_1 \partial y_1} & \cdots & \frac{\partial^2 g}{\partial y_1 \partial y_n} \\ \vdots & & \vdots \\ \frac{\partial^2 g}{\partial y_n \partial y_1} & \cdots & \frac{\partial^2 g}{\partial y_n \partial y_n} \end{bmatrix} \quad \frac{\partial^2 g}{\partial y_k \partial y_j} \underset{\substack{\uparrow \\ \text{Chain} \\ \text{rule}}}{=} \sum_{i=1}^n \frac{\partial}{\partial y_k} \frac{\partial g}{\partial y_j} \underset{\substack{\uparrow \\ \text{Chain} \\ \text{rule}}}{=} \sum_{i=1}^n \frac{\partial}{\partial y_k} \left(\frac{\partial f}{\partial x_i} \right) A_{ij} =$$

$$\underset{\substack{\uparrow \\ \text{Chain} \\ \text{rule}}}{=} \sum_{i=1}^n \sum_{\ell=1}^n \frac{\partial^2 f}{\partial x_\ell \partial x_i} \cdot \frac{\partial x_\ell}{\partial y_k} A_{ij} =$$

$$\underset{\substack{\uparrow \\ \text{Chain} \\ \text{rule}}}{=} \sum_{i=1}^n \sum_{\ell=1}^n \frac{\partial^2 f}{\partial x_\ell \partial x_i} A_{\ell k} A_{ij}$$

Therefore:

$$\nabla^2 g(y) = A^T \nabla^2 f(x) A$$

Now we can relate the Newton's update formulae:

$$\text{For } f: \quad x_{k+1} = x_k - \nabla^2 f(x_k)^{-1} \nabla f(x_k)$$

$$\begin{aligned} \text{For } g: \quad y_{k+1} &= y_k - \nabla^2 g(y_k)^{-1} \nabla g(y_k) = y_k - (A^T \nabla^2 f(x_k) A)^{-1} (A^T \nabla f(x_k)) = \\ &= y_k - A^{-1} (\nabla^2 f(x_k)) (A^T)^{-1} (A^T \nabla f(x_k)) = \\ &= y_k - A^{-1} (\nabla^2 f(x_k)) \nabla f(x_k) \end{aligned}$$

We want to prove that $x_k = Ay_k + b$ for all k . I want to prove it by induction.

We already know that it holds for $k=0$. Let's prove: if $x_k = Ay_k + b \Rightarrow x_{k+1} = Ay_{k+1} + b$.

Multiplying both sides of the Newton's update by A and add b :

$$Ay_{k+1} + b = Ay_k - AA^{-1} (\nabla^2 f(x_k)) \nabla f(x_k) = Ay_k - (\nabla^2 f(x_k)) \nabla f(x_k) = x_{k+1}$$

Then, by induction, $\{x_k\}$ and $\{y_k\}$ are in one-to-one correspondence under the affine transformation $x = Ay + b$.

(b) Under affine transformations, show that the Newton's method with backtracking line search generates the same step length, i.e. the step length s^k of the function $f(x)$ equals to t^k of the function $g(y)$. In addition, show that the newton decrements of both functions are identical. (Bonus (0.5 pt))

1] Backtracking line search selects the step length s_k checking the following condition:

$$f(x_k + s_k \Delta x_k) \leq f(x_k) + \alpha s_k \nabla f(x_k)^T \Delta x_k$$

For $g(y)$ we have:

$$g(y_k + t_k \Delta y_k) \leq g(y_k) + \alpha t_k \nabla g(y_k)^T \Delta y_k$$

From the previous exercise we know that $g(y) = f(Ay + b) = f(x)$, $x_k = Ay_k + b$ and $\Delta x_k = A \Delta y_k$.

$$\text{LHS: } g(y_k + t \Delta y_k) = f(A(y_k + t \Delta y_k) + b) = f(Ay_k + tA\Delta y_k + b) = f(x_k + t \Delta x_k)$$

$$\begin{aligned} \text{RHS: } g(y_k) + \alpha t_k \nabla g(y_k)^T \Delta y_k &= f(x_k) + \alpha t_k (A^T \nabla f(x_k))^T \Delta y_k = f(x_k) + \alpha t_k \nabla f(x_k)^T A \Delta y_k = \\ &= f(x_k) + \alpha t_k \nabla f(x_k)^T \Delta x_k \end{aligned}$$

As we have seen that the Armijo condition is identical for both f and g , the condition will be satisfied at the same time.

So the search will pick the same final step length: $t_k = s_k$

2] The Newton decrement for f is $\lambda_f(x) = (\nabla f(x)^T \nabla^2 f(x)^{-1} \nabla f(x))^{\frac{1}{2}}$

$$\begin{aligned} \text{The Newton decrement for } g \text{ is } \lambda_g(y) &= (\nabla g(y)^T \nabla^2 g(y)^{-1} \nabla g(y))^{\frac{1}{2}} = ((A^T \nabla f(x))^T (A^T \nabla^2 f(x) A)^{-1} (A^T \nabla f(x)))^{\frac{1}{2}} = \\ &= (\nabla f(x)^T A A^{-1} \nabla^2 f(x)^{-1} (A^T)^{-1} A^T \nabla f(x))^{\frac{1}{2}} = (\nabla f(x)^T \nabla^2 f(x)^{-1} \nabla f(x))^{\frac{1}{2}} = \lambda_f(x) \end{aligned}$$

$$\text{So } \boxed{\lambda_f(x) = \lambda_g(y)}$$